



Armada Kalamunda Group

Infection Prevention and Management for Multi-Resistant Organisms Guideline

1. Purpose

This document will provide information to clinical staff on infection prevention and management of multi-resistant organisms (MRO) across all setting within the Armada Kalamunda Group (AKG). Included in this document are the specific management in various clinical areas and screening and clearance requirements.

This document is in accordance with the current state guidelines for [Screening and Management of Multi-resistant Organisms in Healthcare Facilities 2023](#).

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3. Applicability

This document applies to all services and disciplines across the Armadale Kalamunda Group.

4. Consumer engagement

Armadale Kalamunda Group ensures that a patient's micro-alert status will not restrict the provision of care and services.

Newly colonised or infected patients will be informed and given written notification by the Infection Prevention and Management team of their micro-alert status.

5. Guidelines

5.1 Standard precautions

Work practices that constitute the first-line approach to infection prevention and control in the healthcare environment. These are always recommended for the treatment and care of all patients.

5.2 Transmission-based precautions

Extra work practices in situations where standard precautions alone may be insufficient to prevent infection transmission (e.g., for patients known or suspected to be infected or colonised with infectious agents that may not be contained with standard precautions alone).

6. Risk factors for acquiring for MRO infections

- Prolonged length of hospitalisation
- Patients who have invasive procedure/surgery
- Patients with invasive devices (PIVC, PICC, IDC)
- Previous or recent therapy with antimicrobial
- Patients with chronic wounds, discharging wounds and non-intact skin.
- Patients receiving immunosuppressants or with severe chronic diseases.
- Patient requiring full assistance and those not capable of maintaining their personal hygiene.

7. Mode of transmission

Direct Contact	via contaminated hands of healthcare personnel or from colonised patients.
Indirect Contact	via contaminated items, medical equipment, body fluids and surfaces.

8. Micro-alert and description

Public hospitals in Western Australia use a Micro-Alert system to electronically flag an alert through the Web-based Patient Management System (WebPAS) of patients who are colonised or infected with MRO's. The micro alert entered in WebPas will be visible in iSoft Clinical Manager and Enterprise Bed Management System.

Organisms	Micro-Alert	Definition
Methicillin-resistant <i>Staphylococcus aureus</i> (MRSA)	B	Community associated MRSA
	C	Healthcare associated MRSA
	W	Contact of MRSA (Micro C) patient Any person who has shared a patient room with a micro-alert C positive patient prior to the patient having contact precautions initiated and for whom screening has not been undertaken or completed prior to discharge
Vancomycin-resistant Enterococci (VRE)	V	Colonised or infected with VRE
	F	Contact of VRE (Micro V) patient
Carbapenemase Producing Organisms (CPO)	G	Confirmed CPO with presence of a carbapenemase producing enzyme, including but not limited to, KPC, NDM, VIM, OXA and IMP.
	H	Contact of CPO (Micro G) patient Any person who has shared a patient room, bathroom, or toilet facility with a CRE positive patient prior to implementation of contact precautions and for whom screening has not been completed prior to discharge.
<i>Candida auris</i> (<i>C. auris</i>)	J	Colonised or infected with <i>C. auris</i>
	K	Contact of Micro J patient

Multi-resistant gram-negative bacteria (MRGNB – non CRE)	Y	This code is restricted, and it is only assigned at King Edward Memorial Hospital (KEMH), Perth Children's Hospital (PCH) and Fiona Stanley Hospital (FSH) where neonates in high-level intensive care units are at higher risk of a MRGNB bloodstream infection (BSI). Mothers of neonates identified with MRGNB (non-CRE) are also assigned an alert due to their close contact with neonates.
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9. Screening requirements

All patients will be assessed upon admission if they require screening for specific MRO. The following are the recommended screening.

Category of patient	Screening recommendation			
	<i>C.auris</i>	CPO	MRSA	VRE
Repatriation from any overseas hospital or hospitalised for any period, or resided in a Residential Care Facility (RCF) overseas in the last 12 months	✓	✓	✓	✓
Any person who has been hospitalised overnight or resided in a RCF outside of WA in the last 12 months	✗	Only for CPE	✓	✓
Any person who has resided in a RCF within WA in the last 12 months (Any resident from RCF that is screened for MRSA then represent to hospital within two-week period does not require repeat screening for that admission)	✗	✗	✓	✗
Haemodialysis patients (in-centre and satellite) following dialysis outside WA	Only if overseas	Only for CPE	✓	✓
Patient admitted to higher risk ward/unit (ICU)	✗	On admission and either weekly or on discharge from unit		
Contact of known MRO positive case	✓	✓	✓	✓

[Refer to Appendix 3 for MRO specimen collection requirements](#)

10. Inpatient management

Patients will be reviewed by IPM team for micro-alert clearance eligibility. See [appendix 3](#) for clearance criteria and protocol.

Accommodation and Precautions	a. Single room with ensuite or dedicated toilet facility is preferred. If not available, single room and dedicated bathroom facility or single room with dedicated commode. b. Display the appropriate precaution card (Standard Precaution or Contact Precaution) outside patient's room. (Refer to appendix 1 for specific ward management) c. Doors to be always closed or can remain open in certain circumstances. d. Designated non-critical (e.g., BP Machine, stethoscope) or single-patient use equipment should be used wherever possible. e. If reusable items are required, they should remain in the patient's room until discharge or cleaned and disinfected prior to removal from the room. f. Remove all non-essential equipment from the patient's room prior to admission. g. Minimise stocks of disposable equipment kept inside patient's room. Discard all single-use items upon discharge. h. Keep patient notes and bedside chart outside the room. i. Bed and pillow covers must be intact and impermeable. j. If patient placement into a single room is unavailable, cohorting similar micro alerts can only be considered following discussion with the Infection Prevention and Management (IPM) Team and Infectious Diseases (ID) Team. More information on cohorting can be found in the Standard and Transmission Based Procedure.
Personal Protective Equipment (PPE)	Gown a. Should be donned prior to entering patient room if direct patient contact is anticipated. b. Plastic apron is a minimum requirement, however long sleeve fluid resistant gowns are preferred for direct patient care. Gloves a. Assess if gloves are indicated (contact with blood and body fluids). b. Must be available inside the patient room. Protect eyewear / face shields Required if there is a risk of blood or bodily fluid exposure. Face mask Required as per standard precautions. PPE removal

	Removed at the patient door and prior to exiting the room, or in the anteroom if present.
Hand Hygiene	a. Soap and Water – for visibly soiled/contaminated hands b. Alcohol-based Hand Rub – use in accordance with the 5 moments for Hand Hygiene
Room Cleaning	Daily Cleaning - follow hospital procedure for cleaning and disinfection for isolation room. Discharge Clean (once room is vacated) - follow hospital discharge cleaning process for isolation room
Linen Management	a. Change linen daily or immediately if soiled. b. Follow the AKG Linen Management Policy.
Waste Management	As per standard precautions
Catering Staff	a. PPE (gown, apron, gloves) are not required for catering staff who are delivering and removing meal trays. b. Hand hygiene must be performed after leaving the room. c. No additional requirements for washing trays or eating utensils. Disposable crockery/cutlery is not required.
Patient Movement and Transfers	a. Patient movement in the ward should be limited to reduce the risk of MRO transmission. b. Adhere to standard precautions during transfer. c. Notify the receiving unit/facility of Micro-Alert status prior to transfer (internal and external transfers).
Patient Education	a. Educate the patient and family regarding the MRO status. b. Reinforce the importance of hand hygiene. c. On discharge all MRO-positive patients are to be provided with education (verbal and/or written) on the risk of transmission, the importance of notifying healthcare providers of their micro alert status and informed of any clearance procedures if applicable
Visitors	a. PPE is not required for visitors. b. Visitors are advised to perform hand hygiene prior to entering, and leaving, the patient room and following any patient contact.

10. Non-inpatient management

This outlines the management in the non-inpatient units (Not Admitted Overnight), which includes Operating Theatres, Day Procedure Unit, Endoscopy, Radiology, Pathology, Outpatient Clinics and Gyms, Podiatry, Diabetes Clinics.

To prevent the transmission of MROs in the outpatient setting, all staff must apply standard precaution in all aspects of patient care with thorough emphasis on hand hygiene

Precautions	Standard Precautions
PPE	PPE requirement will be based on individual risk assessment
Hand Hygiene	- As per 5 moments for Hand Hygiene - All patients must perform hand hygiene with alcohol-based hand rub or soap and water prior to entering the area.
Cleaning and Disinfecting	- Reusable equipment must be cleaned and disinfected prior to reuse. - Follow the hospital discharge cleaning process after procedure or on discharge.

11. Multi-resistant gram-negative bacteria and extended spectrum beta lactamase

The IPM team will conduct a risk assessment for patient's admitted in the wards and will provide advice on the appropriate precautions to apply.

12. Community visit

This outlines the management of patients in the community who are visited by healthcare workers for their on-going home care.

Standard Precautions	- Healthcare worker conducting home visitation will apply standard precautions. - Healthcare worker should have prior knowledge of the patients' Micro Alert status.
PPE	- PPE requirement will be based on individual risk assessment. - Factors to consider includes: - Discharging wounds and skin exfoliation - Diarrhoea - Indwelling devices - COVID-19 positive cases - Respiratory symptoms - A safe place for the donning and doffing of PPE must be identified. - Safe disposal of used PPE must be ensured
Reusable Items/Equipment	All reusable items and equipment must be cleaned and disinfected after each use. Items/equipment for reprocessing must be placed in a leak proof container and transported to CSSD
Hand hygiene	Perform hand hygiene in accordance with hand hygiene policy

12. Legislative context

The following documents are required to give effect to this guideline:

- [Health Services Act 2016](#)
- [Public Health Act 2016](#)

- [Occupational Safety and Health Act 1984](#)
- Health Department of Western Australia (HDWA) – [Healthcare Associated Infection Surveillance Policy MP 0108/19](#)
- HDWA – [Screening and Management of Multi-resistant Organisms in Healthcare Facilities Policy MP 0177/33](#)
- [East Metropolitan Health Services Infection Prevention and Management Policy 2022](#)

13. Supporting information

The following documents support the implementation of this guideline:

- [Hand Hygiene Policy - AKG-POL-0187-17](#)
- [Linen Management Policy - AKG-PRO-0227-14](#)
- [Cleaning of Reusable Clinical Non-invasive Equipment Procedure - AKG-PRO-0383-17](#)
- [Single Room prioritisation and Cohorting of Inpatients on Transmission Based Precautions Procedure- AKG-PRO-0513/19](#)
- [Patient Screens/Curtains Changing Procedure - AKG-PRO-0350-18](#)
- [Care Worker Immunisation -AKG POL-0188-14-14](#)
- HDWA – [Guidelines for the Screening and Management of Multi-resistant Organisms in Healthcare Facilities](#)
- HDWA – [Guidelines for the Screening and Management of MRSA in Healthcare Workers](#)
- HDWA – [Guidelines for using the Micro-alert system in Western Australian Public Healthcare Facilities](#)

14. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated with.

- Infection Prevention and Management team will conduct daily ward rounds to monitor and review the transmission-based precaution for patients.
- Patient requiring MRO screening are identified and testing will be facilitated.
- Patient placement/accommodation will be reviewed daily to prevent cross infection.
- Standard precaution and transmission-based precaution audits are conducted on a regular basis to monitor compliance.

15. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Partnering with consumers
- Standard 3 – Preventing and Controlling Infection Standards

16. References

1. Australian Guidelines for the Prevention and Control of Infections in Healthcare 2019
2. The national Institute for Occupational Safety and Health – MRSA and the workplace

17. Glossary

The following definitions are relevant to this policy.

Term	Definition
<i>Candida auris</i>	A yeast that is resistant to many antifungal agents, is highly pathogenic and transmissible.
Carbapenemase	A class of enzymes that inactivate carbapenem antibiotics (ertapenem, imipenem or meropenem) with those most identified as: <i>Klebsiella pneumoniae</i> carbapenemase (KPC); New Delhi metallo- β -lactamase (NDM); Verona integron-encoded metallo- β -lactamase (VIM); Oxacillinases (OXA) and Imipenemase (IMP).
Carbapenemase producing organism (CPO)	All organisms identified as producing a carbapenemase.
Colonisation	Is the presence of microorganisms without clinical signs of infection.
Colonisation	Is the presence of micro-organisms without clinical signs of infection.
Contact precautions	A set of infection prevention practices used to prevent transmission of infectious agents that are spread by direct or indirect contact with the patient or the patient's environment which cannot be contained by standard precautions alone. Contact precautions include the use of gloves with an apron or fluid resistant gown (dependant on the degree of risk of contact with blood and body fluids) and other personal protective equipment (PPE) as required as per standard precautions.
Healthcare associated infections	An infection that occurs because of healthcare interventions and may manifest after the patient is discharged.
Healthcare workers	Any person whose activities involve the provision of care either directly or indirectly to patients in a healthcare or laboratory setting and includes those who are employed, honorary, contracted, on student placement or volunteering at the facility. The term is generally applied to all persons working in a healthcare facility (HCF).
Methicillin-resistant <i>Staphylococcus aureus</i> (MRSA)	Those isolates of <i>S. aureus</i> that are resistant to methicillin and consequently all other beta-lactam antibiotics. There are 2 recognised groups of MRSA based on molecular typing: a. Communication associated MRSA (CA-MRSA):

Term	Definition
	These strains of MRSA are identified by molecular typing and have adapted to survive and spread successfully in the community. Few of these strains have been linked to Hospital associated outbreaks due to their epidemic potential. b. Healthcare associated MRSA (HA-MRSA) Refers to distinct stains identified by molecular typing. These strains are known to be highly transmissible within and between HCFs and cause outbreaks. HA-MRSA do not spread efficiently between people in the community.
Micro-alert	A generic term used to refer to the flag applied to the patient medical record on the patient management system in public hospitals to indicate a carrier of or a previously unscreened contact of a person with an MRO
Multi-resistant gram-negative bacteria (MRGNB)	Gram-negative bacteria with resistance to multiple antibiotics. Includes organisms with extended-spectrum beta-lactamase (ESBL) or plasmid-mediated AmpC enzymes as well as multidrug resistant <i>P. aeruginosa</i> and <i>A. baumannii</i> .
Vancomycin resistant enterococci (VRE)	Enterococci are gram-positive cocci that are part of the normal human gut flora. VRE are those isolates of enterococci that are resistant to glycopeptides (vancomycin or teicoplanin).

18. Document contact

Enquiries relating to this guideline may be directed to:

Department: Infection Prevention and Management
 Email: ahsinfection.controlnurses@health.wa.gov.au

19. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V1.2	16/07/2024	13/10/2026	Screening requirement - Repatriation from any overseas hospital or hospitalised for any period or resided in residential care facility overseas in last 12 months

			Updated appendix title to MRO Specimen Collection Requirements and Clearance criteria
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20. Authorisation

This guideline has been authorised and issued by the Director Nursing and Midwifery

Authorised by	Helen Fullarton – Director Nursing and Midwifery
Authorisation date	16/07/2024
Published date	18/07/2024

Appendix 1: Specific ward management

Micro Alert	Area/Ward	Precautions	Management
B (High-risk areas)	ICU	Standard Precautions	- Single room preferably with ensuite or dedicated toilet/bathroom. - Apron/long sleeve gown if risk of contamination of uniform or caring for high-risk patients.
	Colyer		
B (Lower-risk areas)	Canning, MAU, RAC, Campbell, Maud Bellas, Anderton	Standard Precautions	- Patient can be in an open ward or shared room if no risk factors. Single room preferably with ensuite or dedicated toilet/bathroom for high-risk patients. <u>High Risk Patients</u> ▪ Active exfoliative skin conditions (e.g., psoriasis, eczema) ▪ Uncontained discharging wounds ▪ MRSA respiratory tract infection ▪ Inability to comply with personal hygiene whilst in hospital. - Apron/long sleeve gown if risk of contamination of uniform or caring for high-risk patients.
	Renal Dialysis	Standard Precautions	Apron/long sleeve gown if risk of contamination of uniform or caring for high-risk patients.
	Mental Health (in-patient areas)	Standard Precautions	Apron/long sleeve gown if risk of contamination of uniform or caring for high-risk patients.
	Operating theatre, Same day, Endoscopy, Medical Imaging	Standard Precautions	Apron/long sleeve gown if risk of contamination of uniform or caring for high-risk patients.

C	ICU, Canning, MAU, RAC, Colyer, Campbell, Maud Bellas, Anderton	Contact Precautions	Single room preferably with ensuite or dedicated toilet/bathroom.
	Renal Dialysis		Single room or in Micro-Alert C cohort area.
	Mental Health (in-patient areas)	Standard Precautions	- Single room preferably with ensuite or dedicated toilet/bathroom. - If there is risk of contamination of uniform during the provision of care, Apron/long sleeve gown must be worn.
	Operating theatre, Same day, Endoscopy, Medical Imaging		
W	ICU, Canning, MAU, RAC, Colyer, Campbell, Maud Bellas, Mental Health, Anderton	Standard Precautions	- Single room preferably with ensuite or dedicated toilet/bathroom. - If there is risk of contamination of uniform during the provision of care, Apron/long sleeve gown must be worn. - Collect screening swabs for clearance
	All other Areas	Standard Precautions	If there is risk of contamination of uniform during the provision of care, apron/long sleeve gown must be worn. Collect screening swabs for clearance
V	ICU, Canning, MAU, RAC, Colyer, Campbell, Maud Bellas, Mental Health, Anderton	Contact Precautions	- Single room preferably with ensuite or dedicated toilet/bathroom. - If there is risk of contamination of uniform during the provision of care or caring for high-risk patients, apron/long sleeve gown must be worn.
	Renal Dialysis	Contact Precautions	- If there is risk of contamination of uniform during the provision of care or caring for high-risk patients, apron/long sleeve gown must be worn. - Bathroom to be cleaned and disinfected following use
	Emergency Department	Standard Precautions	- Cubicle preferred, if staying greater than 24 hours - If there is risk of contamination of uniform during the provision of care, apron/long sleeve gown must be worn.
	Operating theatres, Same day, Endoscopy, Medical Imaging	Standard Precautions	- If there is risk of contamination of uniform during the provision of care, apron/long sleeve gown must be worn. - Bathroom to be cleaned and disinfected following use

F	All Areas	Standard Precautions	- Single room preferably with ensuite or dedicated toilet/bathroom. - If there is risk of contamination of uniform during the provision of care, apron/long sleeve gown must be worn. - Collect screening swabs for clearance
	ICU, Canning, MAU, RAC, Colyer, Campbell, Maud Bellas, Mental Health, Anderton	Contact Precautions	Single room with ensuite.
	Emergency Department	Standard Precautions	Cubicle preferred, if staying greater than 24 hours If there is risk of contamination of uniform during the provision of care, apron/long sleeve gown must be worn
	Renal Dialysis	Contact Precautions	- Plastic apron as minimum - Long sleeve gown if risk of contamination of uniform or caring for high-risk patients - Bathroom to be cleaned and disinfected following use
J & K	Operating theatres, Same day, Endoscopy, Medical Imaging	Standard Precautions	If there is risk of contamination of uniform during the provision of care, apron/long sleeve gown must be worn.
	ICU, Canning, MAU, RAC, Colyer, Campbell, Maud Bellas, Mental Health, Anderton	Contact Precautions	Single room with ensuite.
	Renal Dialysis	Contact Precautions	- Plastic apron as minimum - Long sleeve gown if risk of contamination of uniform or caring for high-risk patient. - Bathroom to be cleaned and disinfected following use
	Emergency Department	Standard Precautions	- Cubicle preferred, if staying greater than 24 hours. - If there is risk of contamination of uniform during the provision of care, apron/long sleeve gown must be worn
G & H	Operating theatres, Same day, Endoscopy, Medical Imaging	Standard Precautions	- If there is risk of contamination of uniform during the provision of care, apron/long sleeve gown must be worn. - Bathroom to be cleaned and disinfected following use

Appendix 2: Theatre and recovery management for MRO patients

General

- a. There is no IPM requirement for micro-alert patients to be scheduled last on the surgery list of the day.
- b. Patients with micro-alerts are to be managed under “**Standard Precautions**” in theatre and recovery.
- c. Staff should adhere to the 5 moments for hand hygiene, including the appropriate use of gloves when there is a risk of contact with body fluids.
- d. Disposable aprons or gowns must be worn if there is a risk of contaminating scrubs/uniform.
- e. Patient can be placed in theatre holding bay under standard precautions.
- f. Anaesthetic circuits replaced, and anaesthetic machine cleaned in between each case as per routine.
- g. Remove unnecessary equipment that is not needed during the surgery.

Ward/Theatre Transfers

- a. Standard precautions will apply during transfer to and from the ward.
- b. Transport staff should strictly adhere to hand-hygiene practices.
- c. Wearing of gloves is not required during transport.

Post-operative/Recovery Management

If patient is transferred to another area after the procedure, the receiving unit must be notified of the Micro-Alert status prior to transfer.

- a. Patient can be placed in the Recovery Area/Room post-procedure.
- b. Standard precautions are applied throughout the recovery period.
- c. Strict adherence to 5 moments of hand hygiene must be observed.
- d. PPE (gown and gloves) use as per risk assessment for contact with body fluids.
- e. Shared patient equipment must be cleaned and disinfected between each patient as per standard precaution requirements.
- f. If patient is admitted overnight, contact precautions shall be applied in addition to standard precaution.

Cleaning and Disinfection (Theatre and Recovery)

- a. Standard routine clean and disinfection must be performed after the procedure/recovery (**one step clean with a combined detergent and disinfectant product – such as Chlor-clean**). This is required between all theatre cases, regardless of micro alert status.
- b. Clean and disinfect all surfaces touched by staff or the patient including mattresses, fixtures, equipment, furniture, monitoring equipment, machinery, control panels, monitor knobs, work areas, computers, and handles.
- c. For significant gross soilage with blood or body fluids, a two-step clean and disinfection of all contaminated surfaces and equipment is required as per the AKG Blood and Body Fluid Spills Management Procedure (AKG-PRO-0061-18).

Two step clean:

Step one – mechanically remove organic matters (body fluids/Faeces) with detergent/neutral cleaner (**Taski -R2 Room Care**). Depending on the size of the spill a spill kit may be required for this step.

Step two – wipe all contaminated surfaces and equipment with disinfectant product (**Chlor clean**) - Allow the product to dry thoroughly. (Contact time required for Chlor-clean is about 5minutes)

Appendix 3: MRO specimen collection requirements and clearance criteria

Screening Samples for MROs			
Additional samples are required from the following sites if present for all MRO screening: - any wounds, ulcers, skin lesions or wound drain fluid - any invasive device sites e.g., PEG, tracheostomy, intravascular device - urine specimen from indwelling or suprapubic urinary catheters - endotracheal aspirate Note: Patients with an enterostomy must have a stomal swab collected instead of a rectal swab			
MRO	Alert	Criteria for clearance	Swabs to collect
C. auris	J	No clearance screening and repeated screening not required	Not applicable
	K or screening as per section 9-screening requirement	Clear when all 3 sets of screening results are final	Collect samples on 3 consecutive days. A single swab i.e a composite swab 1 swab-4 sites, without transport medium. Sample both axillae, then both sides of the groin by rubbing the swab tip firmly back and forth in skin crease, 3-5 times for each site. Swab can be pre-moistened with sterile water. Use of a double headed swab will increase the sensitivity.
		NB: At risk patients should ideally be screened while off anti-fungal medications for at least 7 days and not within 48 hours of using antiseptic body washes as these treatments may result in false negative screens.	
CPO	G	No clearance screening and repeated screening not required	Not applicable
	H	Clear on receipt of 3 sets of negative results	3 rectal swabs (or stomal) pre-moistened with sterile water or faecal specimen collected on 3 consecutive days following the last known contact
	screening as per section 9-screening requirement	Clear on receipt of negative results	A single rectal (or stomal) swab pre-moistened with sterile water or a faecal specimen, separate to VRE swab.
MRSA	B or C	Can only be obtained 3 months after last known positive result. Person must not have used any topical antiseptics for the past week and is not on antibiotics at time of screening. Clear on receipt of 2 sets negative results.	Two sets of nasal and two sets pharyngeal swabs pre-moistened with sterile water. Sets can be taken consecutively on the same day.
	W	Clear on receipt of negative results.	One set of nasal and pharyngeal swabs pre-moistened with sterile water.
	screening as per section 9-screening requirement	Clear on receipt of negative results	One set of nasal and pharyngeal swabs pre-moistened with sterile water. The pharyngeal swab is particularly important if decolonisation is planned. An umbilical swab is required from neonates.

MRO	Alert	Criteria for clearance	Swabs to collect
VRE	V	Not to be cleared within one year of a positive result.	Not applicable
		If four years or more since last positive specimen, then no specimens are required. Micro-alert can be cleared.	No specimens required.
		If between one and four years since last positive specimen further screening required. On receipt of negative results micro-alert can be cleared.	3 rectal swabs (or stomal) pre-moistened with sterile water or a faecal specimen collected on 3 consecutive days following the last known contact
		Note: There is currently no VRE clearance process for haemodialysis patients.	
	F	Unscreened contact	3 rectal swabs (or stomal) pre-moistened with sterile water or a faecal specimen collected on 3 consecutive days following the last known contact
	screening as per section 9-screening requirement	Clear on receipt of negative result	single rectal swabs (or stomal) pre-moistened with sterile water or a faecal specimen separate to CPO swab.
ESBL/GRE	Y	Currently there is no clearance screening protocol for micro-alert Y.	

Sample Collection Kit	
Charcoal Swab (Copan)	Use for MRSA, VRE and CPO
Single Headed Dry Swab (Copan)	Use for <i>C. auris</i>

Specimen Collection Procedure
- Any swabs collected from a dry site e.g. nostrils or non-discharging lesions are to be pre-moistened with sterile water or normal saline. - Swabs collected from moist sites e.g. discharging wounds, do not need to be pre-moistened. Don PPE as per standard precautions when collecting specimen
Axilla Swab - Rub both sides of the swab tip over each axilla skin surface, swiping back and forth 5 times per axilla (<i>targetting the crease in the skin where the arm meets the body</i>). - Return the swab to the swab container, avoiding contamination.
Groin - Rub both sides of the swab tip on the skin surface of each side of the region, swiping back and forth 5 times each side (<i>targeting the inguinal crease in the skin where the leg meets the pelvic region</i>). - Return the swab to the swab container, avoiding contamination.

Nasal

- Rotate a single swab, 2-3 times around the insided of the nostrils, using the same swab for both nostrils.
- Return the swab to the swab container, avoiding contamination.

Pharyngeal/throat

- Swab the posterior pharynx and lateral walls of the pharynx i.e. 'tonsilar' area without touching the buccal mucosa or tongue.
- Return the swab to the swab container, avoiding contamination.

Rectal

- Insert swab 1cm into rectum and gently rotate 360 degrees.
- For patients with enterostomies a stomal specimen is required.
- Faeces must be evident on these swabs
- Return the swab to the swab container, avoiding contamination.

Appendix 4: MRSA screening requirements and management for Healthcare Works (HCW's), including students, trainees, and volunteers

MRSA screening requirements for HCW

On commencement of employment	▪ All HCWs who will have clinical contact, are required to have their MRSA risk status assessed as part of a pre-employment/commencement process. ▪ MRSA surveillance screening is required prior to commencement of work if the HCW: ○ has worked in a healthcare facility or RCF outside of WA in the previous 12 months including volunteer placements, or ○ has been hospitalised outside of WA in the previous 12 months
Current Staff	▪ Current HCWs who perform clinical duties in a healthcare facility or RCF outside of WA, including volunteer placements, and are returning to work, are to have MRSA surveillance screening performed on their return. ▪ HCWs can continue clinical duties pending results.
Visiting HCWs	▪ Any visiting HCWs who will be involved in clinical contact for a period greater than five days, require screening prior to placement.
Additional HCW screening	▪ Screening of HCWs should be considered when a single strain outbreak continues despite adherence to infection prevention and control measures

Obtaining samples for MRSA screening

Specimen site	Nostrils, Throat and any skin lesions or wounds
Method of collection	For Nostrils: • Pre-moistened a single swab with sterile normal saline or sterile water. • Insert swab into the nostril and rotate 2-3 times (the same swab used for both nostrils). For Throat: • Swab the posterior pharynx and lateral walls of the pharynx (tonsillar area) without touching the buccal area or tongue. For wounds • Swab any discharging wounds, ulcers, or skin lesions. • Pre-moistened swab with normal saline or sterile water for non-discharging wounds, ulcers, or lesions.

Appendix 5: MRSA decolonisation and treatment

Management of HCW colonised with MRSA

For the management of HCW colonised with MRSA refer to:

- [MRSA Screening and Management of Healthcare Workers](#)
- [MRSA decolonisation treatment information for healthcare provider](#)
- [AKG Health Care Worker Immunisation Policy](#)
- [MRSA decolonisation treatment instructions for healthcare workers](#)

Decolonisation treatment for patients with MRSA

- For decolonisation treatment for patients, refer to the [MRSA decolonisation treatment – information for consumers](#)